

5(a)	any four from: 1 geographical isolation / barrier qualified ; Ignore separation 2 no, interbreeding / gene flow (between European and American populations) ; Ignore reproductive isolation here (needs to be late on in process) 3 different, environmental (conditions) / selection pressures ; 4 random / different, mutations ; 5 different, alleles selected for / gene pool / changes in allele frequency ; 6 populations have different, morphological / physiological / behavioural, features ; 7 after, a long time / many generations, reproductive isolation occurs / cannot interbreed to produce fertile offspring ; 8 allopatric (speciation) ;	4
------	---	----------

Question	Answer	Marks
5(b)	any three from: 1 not possible / difficult, to come to a (reliable) conclusion ; A AW 2 small difference in height / heights are similar (so natural selection not more important than genetic drift) ; 3 taller / lighter, bison could be an advantage in reaching higher vegetation (so natural selection may be important) ; 4 difference in mass, may not be due to genetic factors / may be due to nutrient value of diet / may be due to quantity of food, (so natural selection may not be important) ; 5 genetic drift (in ancestral population) may have led to, European bison being, taller / lighter or American bison being, shorter / heavier ;	3
5(c)	1 European bison has a, close / recent / common, ancestor with that of, cattle / <i>Bos</i> or European bison is more closely related to, cattle / <i>Bos</i>, (than to American bison) ; 2 European bison may have interbred with, cattle / <i>Bos</i> ; 3 mitochondria are inherited from, mother / female ;	3

Question	Answer	Marks
5(d)	<i>any four from:</i> 1 obtain / use, blood / cells / tissue / DNA ; 2 from, both species / cattle and European bison ; 3 use PCR to, replicate / amplify, DNA ; 4 use, electrophoresis (described) / DNA profiling / DNA fingerprinting, to see genetic differences / compare the banding (patterns) ; 5 sequence DNA and, count / identify, (percentage) difference / similarity or sequence DNA and compare the, nucleotide / base, sequence ; 6 use, genome / DNA, microarray ; R for gene expression 7 <i>ref. to</i> bioinformatics / database / (computer) software ; 8 similarity indicates, close relationship / more recent split ; 9 if they can breed to produce (viable) offspring (they are closely related) ;	4